# Canon Expectable Signals

F.G. Ersezer, *Student Member, IEEE*

***Abstract*—In this paper, we study the Q-Transformation, a sample-and-hold uniform quantizer with amplitude threshold $\mu$ and time threshold $\tau$, and characterize the expectable signals that it leaves unchanged. We call these canon expectable signals and show that they are exactly the signals that are constant on a $\tau$-grid and take values in $\mu\mathbb{Z}$. Under the explicit hypothesis that physical signals are canon for some fundamental thresholds $(\mu_1, \tau_1)$, we prove that a practical filter with thresholds $(\mu_2, \tau_2)$ composes consistently with the fundamental quantization if and only if $\mu_2/\mu_1$ and $\tau_2/\tau_1$ are positive integers; that the quantized error over any grid-aligned interval is an exact integer multiple of $\mu_1^2\tau_1$; and that the signal-to-noise ratio is rational whenever the threshold ratios are rational. We also derive an exact finite-lattice correction to the classical high-resolution error formula. The results hold without any physical assumption for requantization of signals that are already digital.**

***Index Terms*—Quantization, quantization noise, sample-and-hold, lattice signals**

## I. Introduction

In signal processing, as a stochastic process, the transformation and quantization of signals are fundamental operations that enable the conversion of continuous signals into discrete forms for analysis, storage, and transmission. Traditional quantization methods, such as uniform scalar quantization and vector quantization, primarily focus on efficient sample compression by minimizing the quantization noise. Their analysis usually treats the input as continuous-valued and the quantization error as approximately uniform noise [1], [12], [16].

This paper asks the reverse question: which signals does a quantizer leave exactly unchanged, and what follows if the input itself already lives on a grid? We term these signals *canon expectable signals*. The question has two motivations. The first is physical and speculative: several proposals suggest that measurable quantities may have smallest units [9], [20]. The second is practical and certain: any signal that has already been digitized lives on a grid, and requantizing it (bit-depth reduction, decimation by an integer factor with hold) is a grid-to-grid map.

The Canon Q-Transformation builds on the foundation of classical quantization techniques and on measurement limits in physical reality [9]. Every result below states its assumptions, and the physical hypothesis is kept separate from the mathematics. All theorems also hold in the digital-requantization reading, where no physical assumption is needed.

Revised version, 31 October 2026; this version corrects the error and SNR derivations of the first version and states its physical assumption explicitly.

The contributions of this paper are fourfold:

1) a precise definition of the Q-Transformation as a sample-and-hold uniform quantizer, with an explicit convergence bound;
2) a characterization of canon expectable signals as lattice signals, with a canonical basis of box functions;
3) a necessary and sufficient condition for a practical filter to compose consistently with the fundamental quantization: integer threshold ratios;
4) exact expressions for the quantized error and the SNR as integer counts, rationality of the SNR under rational threshold ratios, and a finite-lattice correction to the classical error formula.

This paper contains no experiments and makes no claim of improved fidelity over existing quantizers; Section IV states what the framework does and does not show.

## II. Literature Review

Signal processing, as a general concept, is a stochastic process [10] that begins with an input as an expected signal and a transformation [11] of the input into an output by a certain measurement modulation. Quantization (measurement modulation of input) is generally the crucial process in signal processing, transforming continuous (or discrete) signals into another (or the same) set of discrete values. The main practical track of quantization is related to filters, and generalizations of filters can be made into various fields of physics such as optics and electronics [8].

The main theoretical concepts related to the spectral characteristics of quantized signals, including the trade-offs between signal distortion and quantization levels, go back to the early work of Bennett [1]. In Bennett's model, a uniform quantizer with step $\Delta$ adds error that is approximately uniform, with mean square $\Delta^2/12$ for rounding. Lloyd [2] and Max [17] derived optimal non-uniform scalar quantizers for a known input distribution, and Widrow and Kollár [16] give a modern treatment of when the uniform-noise model is exact.

Vector quantization quantizes blocks of samples jointly. Its standard design algorithm is due to Linde, Buzo and Gray [18], and the field is reviewed in Gray's tutorial [21], the survey of Gray and Neuhoff [3], and the book of Gersho and Gray [13]. Zamir and Feder [4], [5] studied lattice quantization noise, proposing structured quantization schemes that achieve lower distortion by leveraging the geometric properties of lattices. More recently, Agrell and Allen [6] investigated optimal lattice quantizers. For

a related theoretical aspect see [14].

The Q-Transformation studied here is none of these: it is a uniform scalar quantizer preceded by sample-and-hold. Its contribution lies in the analysis of the signals it leaves unchanged, not in the quantizer itself. We also note that the name "Q-Transform" commonly denotes the constant-Q spectral transform [19], which is unrelated; we keep the name for continuity with the first version of this paper. Finally, proposals for a minimal length or time in physics are reviewed in [20]. None is established, and SI base units are defined through fixed constants [9] rather than through smallest measurable increments. We therefore treat fundamental thresholds as a hypothesis.

## III. Methodology

### *Theoretical Foundations*

The Q-Transformation begins with the representation of the expected signal $x(t)$ over a continuous time interval $t \geq 0$. The primary goal is to transform this signal into a quantized version $y(t)$.

#### *1) Non-Canon Expected Signal Representation*

The non-canon expected signal $x(t)$ can be represented as a series expansion:

$$x(t) = \sum_{i=1}^{n} a_i \, \phi_i(t)$$

where $a_i$ are the coefficients and $\phi_i(t)$ are the basis functions.

#### *2) Q-Transform Framework*

The transformation of the non-canon expected signal into its quantized form $y(t)$ is defined as:

$$y(t) = QT\{x(t)\}$$

Here, $QT\{\cdot\}$ denotes the Q-Transformation process, which quantizes the signal with an amplitude threshold $\mu$ and a time threshold $\tau$.

#### *3) Q-Transform Equations*

The transformation equations (without the delay of filter) are defined as follows:

$$QT_{+}^{+}\{x(t)\} = \mu \cdot \left\lfloor \frac{1}{\mu} \cdot x\Big(\tau \cdot \Big\lfloor \frac{t}{\tau} \Big\rfloor\Big) \right\rfloor$$

$$QT_{-}^{+}\{x(t)\} = \mu \cdot \left\lceil \frac{1}{\mu} \cdot x\Big(\tau \cdot \Big\lfloor \frac{t}{\tau} \Big\rfloor\Big) \right\rceil$$

$$QT_{+}^{-}\{x(t)\} = \mu \cdot \left\lfloor \frac{1}{\mu} \cdot x\Big(\tau \cdot \Big\lceil \frac{t}{\tau} \Big\rceil\Big) \right\rfloor$$

$$QT_{-}^{-}\{x(t)\} = \mu \cdot \left\lceil \frac{1}{\mu} \cdot x\Big(\tau \cdot \Big\lceil \frac{t}{\tau} \Big\rceil\Big) \right\rceil$$

Where;

$$\tau, \mu \in \mathbb{R}_{>0}$$

Each equation is a sample-and-hold in time (the inner floor or ceiling) followed by uniform rounding in amplitude (the outer floor or ceiling). We state results for the causal floor variant $QT_{+}^{+}$ and write $QT_{\mu,\tau}$ for it; the other three behave the same way up to sign conventions and cell endpoints, which are sets of measure zero.

The Q-Transformation is applied to the signal itself. It is not linear: in general $QT\{x+z\} \neq QT\{x\} + QT\{z\}$ and $QT\{a\,x\} \neq a\,QT\{x\}$. Applying it to each basis function $\phi_i$ and recombining therefore does not give $QT\{x\}$, and the sum $\sum a_i \, QT\{\phi_i\}$ need not lie on the $\mu$-grid unless every $a_i$ is an integer.

### *Generalization on the Limits*

Limits on the Q-Transform separate the distinct modes of expected signals. Such cases are obtained when the measurement limitations of the Q-Transformation are infinitely small:

$$\lim_{\mu,\tau \to 0} QT_{\mu,\tau}\{x(t)\}$$

Let $x$ be bounded on $[0,T]$ and let $\omega_x(\tau) = \sup\{|x(s) - x(s')| : |s - s'| \leq \tau\}$ be its modulus of continuity there.

*Proposition 1:* For every $t \in [0,T]$,

$$\big| x(t) - QT_{\mu,\tau}\{x(t)\} \big| \leq \mu + \omega_x(\tau)$$

*Proof:* The sample point $\tau\lfloor t/\tau \rfloor$ lies within $\tau$ of $t$, so the hold changes the value by at most $\omega_x(\tau)$; floor rounding then changes it by a value in $[0,\mu)$. ■

Hence, when the measurement limitations of the Q-Transform are infinitely small, the Q-Transformation of expected signals (as output) acts as the expected signals (as input):

$$x(t) = \lim_{\mu,\tau \to 0} QT_{\mu,\tau}\{x(t)\}$$

The limit is uniform when $x$ is uniformly continuous and holds at every continuity point of $x$, hence almost everywhere for Riemann-integrable signals. It can fail at discontinuities, so it does not hold for every signal. It holds regardless of whether the signal itself is canon or non-canon.

### *Fundamental Measurement Limit Considerations*

The canon track of signals concerns thresholds $\mu_1$ and $\tau_1$ that may exist in physical reality. Such limitations could be represented as a smallest unit of time, force, energy, and so on.

*Hypothesis H (physical, unproven):* For a given physical quantity there exist fundamental thresholds $\mu_1, \tau_1 > 0$ such that every physical signal of that quantity is a canon expectable signal with respect to $(\mu_1, \tau_1)$.

No such thresholds are established for time, force or energy [20], so Hypothesis H is an assumption, and every result below that

uses it says so. In the digital reading, $\mu_1$ and $\tau_1$ are the amplitude step and sample period of an existing digital signal, and H holds by construction.

*1) Canon Expectable Signal Representation*

Canon expectable signals must satisfy:

$$x(t) = QT_{\mu_1,\tau_1}\{x(t)\}$$

This expression is the "Canon Q-Transformation": $x(t)$ is equal to its own quantized transformation when $\mu = \mu_1$ and $\tau = \tau_1$. Let $\beta_k(t)$ be the indicator function of the cell $[k\tau_1, (k+1)\tau_1)$.

*Proposition 2:* A signal satisfies the Canon Q-Transformation if and only if it can be written as

$$x(t) = \sum_{k\geq 0} \mu_1 m_k \beta_k(t), \qquad m_k \in \mathbb{Z}$$

*Proof:* The output of $QT_{\mu_1,\tau_1}$ is always of this form, since the sample point $\tau_1 \lfloor t/\tau_1 \rfloor$ is constant on each cell and floor rounding produces values in $\mu_1\mathbb{Z}$; so every fixed point has this form. Conversely, such an $x$ is constant on each cell, so the hold leaves it unchanged, and floor rounding leaves values in $\mu_1\mathbb{Z}$ unchanged. ■

The canonical basis of canon expectable signals is therefore the box functions $\beta_k$, with coefficients that are integer multiples of $\mu_1$. These basis functions satisfy the Canon Q-Transformation themselves, while smooth bases such as sinusoids or polynomials do not.

*2) Canon Q-Transform Framework*

The transformation of the canon expectable signal into its quantized form $y(t)$ by a non-perfect practical filter with thresholds $\mu_2 \geq \mu_1$ and $\tau_2 \geq \tau_1$ is:

$$y(t) = QT_{\mu_2,\tau_2}\{x(t)\}$$

Under Hypothesis H, $x$ is itself the fundamental quantization of the underlying signal, so the measurement chain can also be written as $QT_{\mu_2,\tau_2}\{QT_{\mu_1,\tau_1}\{x(t)\}\}$. The following theorem gives the exact condition under which the two descriptions agree for every signal.

*Theorem 1:* For all bounded signals $x$,

$$QT_{\mu_2,\tau_2}\{QT_{\mu_1,\tau_1}\{x(t)\}\} = QT_{\mu_2,\tau_2}\{x(t)\}$$

if and only if

$$\mu_1 \cdot n_1 = \mu_2 \ \text{ and } \ \tau_1 \cdot n_2 = \tau_2, \qquad n_1, n_2 \in \mathbb{N}^+$$

*Proof (sufficiency):* Rounding acts pointwise and hold only relabels time, so they commute. In time, $\tau_2 \lfloor t/\tau_2 \rfloor$ is a multiple of $\tau_1$, so holding at $\tau_1$ and then at $\tau_2$ equals holding at $\tau_2$. In amplitude, the identity $\lfloor \lfloor u \rfloor / n \rfloor = \lfloor u/n \rfloor$ for $n \in \mathbb{N}^+$, applied with $u = v/\mu_1$, shows that rounding to $\mu_1$ and then to $\mu_2$ equals rounding to $\mu_2$.

*Proof (necessity):* If $r = \mu_2/\mu_1 \notin \mathbb{N}^+$, take $x \equiv \mu_2$. Directly the output is $\mu_2$; through $\mu_1$ the value becomes $\mu_1 \lfloor r \rfloor < \mu_2$, which then rounds to 0. If $s = \tau_2/\tau_1 \notin \mathbb{N}^+$, take $x(t) = \mu_2$ for $t \geq \tau_2$ and 0 otherwise. At $t = \tau_2$ the direct output is $\mu_2$, while the composite samples $x$ at $\tau_1 \lfloor s \rfloor < \tau_2$ and outputs 0. ■

A filter with non-integer threshold ratios is still well defined; it simply does not agree with the fundamental quantization in this sense. In the digital reading, Theorem 1 is the familiar case of bit-depth reduction by a power of two (24 to 16 bits is a ratio of $2^8$) and decimation by an integer factor. Two consequences follow under Hypothesis H and integer ratios.

*Perfect filter:* When $\mu_2 = \mu_1$ and $\tau_2 = \tau_1$, the filter output equals the input (expected) signal. Such filters are "perfect" in the sense of noise configuration on the practical side.

*Two filters:* If the thresholds $(\mu_2, \tau_2)$ of one filter and $(\mu_3, \tau_3)$ of another are both integer multiples of $(\mu_1, \tau_1)$, then

$$\frac{\mu_2}{\mu_3} \in \mathbb{Q} \ \text{ and } \ \frac{\tau_2}{\tau_3} \in \mathbb{Q}$$

*Metrics of Evaluation*

Throughout this subsection we assume Hypothesis H, so $x = \sum \mu_1 m_k \beta_k$; we integrate over $[0, T)$ with $T = K\tau_1$; and $y = QT_{\mu_2,\tau_2}\{x\}$ with $\mu_2 = n_1\mu_1$ and $\tau_2 = n_2\tau_1$ unless stated otherwise.

*1) Quantized Error*

The quantized error is the integrated squared difference between the original signal and the quantized signal [2], [12], [13], [15]:

$$E = \int_0^T \left(x(t) - QT_{\mu_2,\tau_2}\{x(t)\}\right)^2 dt$$

By Theorem 1, $y$ is itself a canon expectable signal, so the error $e = x - y$ is one too: $e = \sum \mu_1 d_k \beta_k$ with $d_k \in \mathbb{Z}$.

*Proposition 3:* The original expectable signal and the quantized error integrate as

$$\int_0^T x(t)^2\, dt = \mu_1^2\, \tau_1 \sum_{k=0}^{K-1} m_k^2$$

$$E = \mu_1^2\, \tau_1 \sum_{k=0}^{K-1} d_k^2$$

*Proof:* Each function is constant on cells of length $\tau_1$, so each integral is $\tau_1$ times a sum of squared cell values. ■

This outcome is reasoned with the countableness of quantization: the integrals are exact integer multiples of the unit $\mu_1^2\tau_1$. Note that $E$ is not $\int x^2\, dt - \int y^2\, dt$, because $\int (x-y)^2 = \int x^2 - 2\int xy + \int y^2$. For example, with $\mu_1 = \tau_1 = 1$, $m = (3,5)$ and $\mu_2 = 2$, the output is $y = (2,4)$, the error is $e = (1,1)$ and $E = 2$, while $\int x^2 - \int y^2 = 34 - 20 = 14$.

*Proposition 4:* Let $\tau_2 = \tau_1$ and $n = n_1$. Then each $d_k \in \{0,1,\ldots,n-1\}$, so $E \le (n-1)^2\mu_1^2 T < \mu_2^2 T$. If, in addition, the residues $m_k$ mod $n$ are equidistributed over the $K$ cells, then

$$\frac{E}{T} = \mu_2^2 \frac{(n-1)(2n-1)}{6n^2} \xrightarrow[n\to\infty]{} \frac{\mu_2^2}{3}$$

*Proof:* Floor rounding leaves a remainder in $[0,\mu_2)$ on the $\mu_1$-grid. Under equidistribution, $d_k$ takes each value $0,\ldots,n-1$ equally often, and the mean of $d^2$ over those values is $(n-1)(2n-1)/6$. ■

The limit $\mu_2^2/3$ is the classical mean squared error of a floor quantizer under the uniform-error model [1], [16]; Proposition 4 is its exact finite-lattice version. The ratio to the classical value is 0.375 for $n = 2$ and about 0.91 for $n = 16$. The same argument for a rounding quantizer with $n$ odd gives $\mu_2^2(1-1/n^2)/12$, the finite-lattice correction to Bennett's $\mu_2^2/12$.

*2) Signal-to-Noise Ratio (SNR)*

The SNR is the ratio of the signal power to the noise power [12]:

$$SNR = \frac{P_{signal}}{P_{noise}}$$

where the (original) signal power over the interval $T$ is

$$P_{signal} = \int_0^T |x(t)|^2\, dt$$

and the (quantized) noise power over the interval $T$ is

$$P_{noise} = \int_0^T \left| x(t) - QT_{\mu_2,\tau_2}\{x(t)\} \right|^2 dt$$

In decibels:

$$SNR\,(dB) = 10\log_{10}\left(\frac{P_{signal}}{P_{noise}}\right)$$

*Theorem 2:* Assume Hypothesis H, and let $\mu_2/\mu_1 = p/q$ and $\tau_2/\tau_1 = r/s$ be rational, not necessarily integers. Let $\theta = \tau_1/s$ and let $T$ be a multiple of $\theta$. If $P_{noise} > 0$, then

$$SNR\,(dB) = 10\log_{10}(A), \qquad A = \frac{N_S}{N_E} \in \mathbb{Q}, \quad N_S, N_E \in \mathbb{N}^+$$

*Proof:* All values of $x$ and $y$ lie in $(\mu_1/q)\mathbb{Z}$, and both signals are constant on cells of length $\theta$. Each integral is therefore $(\mu_1/q)^2\theta$ times a non-negative integer, and the common factor cancels in the ratio. ■

Rational threshold ratios suffice; integer ratios are not needed for this result. Q-Transformations alone do not need to satisfy this identity, since without Hypothesis H the constant $A$ can be an element of $\mathbb{R}\setminus\mathbb{Q}$. For example, for $x \equiv \sqrt{2}$ and $\mu_2 = 1$, $y \equiv 1$ and $A = 2/(\sqrt{2}-1)^2 = 6+4\sqrt{2}$. If $P_{noise} = 0$ (the perfect filter), the SNR is infinite.

## IV. Conclusion

Canon expectable signals, the fixed points of the Q-Transformation, are exactly the lattice signals. On a lattice, a practical filter agrees with the fundamental quantization if and only if its threshold ratios are integers, and the quantized error and the SNR become exact integer counts. These results hold unconditionally for requantization of digital signals, where Proposition 4 gives exact finite-$n$ corrections to classical error formulas that matter for small $n$.

The framework changes the signal model, not the quantizer, and it does not by itself reduce quantization noise. Applied to physical reality, the results rest on Hypothesis H, which is unproven; under Planck-scale candidates the ratios $n_1, n_2$ of any real instrument would be astronomically large and the corrections negligible. Rationality of the SNR cannot be tested experimentally, since the rationals are dense in the reals. Future work includes simulations comparing Proposition 4 with measured requantization error on audio and image data, extension to rounding and dithered quantizers, and conditions under which the equidistribution assumption holds.

## Data Availability Statement

Some of the data is available to the public, while some of the data has prerequisites, and some of the data is located at an outdated website. The author has given information about the contents he has read below. All referenced pages are at the equivalent of the internet domain's reference pages (logical), not the original book's or reference's pages (physical). References [16]–[21] were added in this revision.


[1] *Spectra of Quantized Signals*, W. R. Bennett: 10.1002/j.1538-7305.1948.tb01340.x

[2] *Least Squares Quantization in PCM*, Stuart P. Lloyd: 10.1109/TIT.1982.1056489

[3] *Quantization*, Robert M. Gray, David L. Neuhoff: 10.1109/18.720541

[4] *On lattice quantization noise*, R. Zamir, M. Feder: 10.1109/DCC.1994.305946

[5] *On universal quantization by randomized uniform/lattice quantizers*, R. Zamir, M. Feder: 10.1109/18.119699

[6] *On the Best Lattice Quantizers*, Erik Agrell, Bruce Allen: arXiv:2202.09605

[7] *The theory of signal detectability*, W. Peterson, T. Birdsall, W. Fox: 10.1109/TIT.1954.1057460

[8] *Emerging Trends in Techniques and Technology as Applied to Filter Design*, Richard V. Snyder, Giuseppe Macchiarella, Simone Bastioli, Cristiano Tomassoni: 10.1109/JMW.2020.3028643

[9] *The role of fundamental constants in the International System of Units (SI): present and future*, B. N. Taylor, P. J. Mohr: 10.1109/19.918192

[10] *The stochastic filtering problem: a brief historical account*, Dan

Crisan: 10.1239/jap/1417528463

[11] *On the optimal filtering of diffusion processes*, Moshe Zakai: 10.1007/BF00536382

[12] *Digital Coding of Waveforms: Principles and Applications to Speech and Video*, N. S. Jayant, Peter Noll. Prentice-Hall, Inc., 1984.

[13] *Vector Quantization and Signal Compression*, Allen Gersho, Robert M. Gray. Kluwer, 1992.

[14] *On Existence of Digital Measure Constant*, F. G. Ersezer: 10.36227/techrxiv.24417994.v2

[15] *A Unifying View of Second-Order Measures of Quality for Signal Classification*, William A. Gardner: 10.1109/TCOM.1980.1094735

[16] *Quantization Noise: Roundoff Error in Digital Computation, Signal Processing, Control, and Communications*, Bernard Widrow, István Kollár. Cambridge University Press, 2008.

[17] *Quantizing for minimum distortion*, Joel Max. IRE Trans. Inf. Theory, vol. 6, no. 1, pp. 7–12, 1960.

[18] *An algorithm for vector quantizer design*, Y. Linde, A. Buzo, R. M. Gray. IEEE Trans. Commun., vol. 28, no. 1, pp. 84–95, 1980.

[19] *Calculation of a constant Q spectral transform*, Judith C. Brown. J. Acoust. Soc. Am., vol. 89, no. 1, pp. 425–434, 1991.

[20] *Minimal length scale scenarios for quantum gravity*, Sabine Hossenfelder. Living Rev. Relativ., vol. 16, art. 2, 2013.

[21] *Vector quantization*, Robert M. Gray. IEEE ASSP Magazine, vol. 1, no. 2, pp. 4–29, 1984.